A golden EC benchmark battery

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The original field-and-curve report is extended here with executable polynomial relation-search fixtures, measured solver results, operational isogenies, two-term endomorphism decompositions, and a separate genuine GLV/GLS four-term example. This addendum gives the current artifact counts and measurement contract; these counts supersede any earlier vector inventory.

Layer	Current content	Purpose
Arithmetic corpus	7 fields; 56 curves; 6,931 known answers	Arithmetic, encoding, rejection, subgroup membership, maps
Polynomial battery	69 fixtures; main battery 55 (49 binary, 6 prime)	Two-factor Semaev relation solving with exact finite domains
GLV2 / Frobenius	30 supported of 56; 480 scalar vectors	Map stability, eigenvalues, lattices, signed reconstruction
GLV + GLS4 supplement	82-bit quadratic field; 80-bit subgroup	Verified four-term lattice with 20-bit component bound

Which curves can be grouped?

Over each fixed field, equal Frobenius trace characterizes an isogeny class; the 56 represented models occupy 52 such classes. Equal j-invariant groups geometric isomorphism classes and twists, and is a different partition. An ordinary endomorphism-order discriminant records further structure inside an isogeny class. None of these labels determines the performance of a particular polynomial presentation.

Fields retained from the original report

Prime fields: $2^{61}-1$, $2^{64}-2^{32}+1$, $2^{83}-2^{59}-1$, and $2^{83}-55$. Binary fields: degrees 53, 79, and 83 with the explicit irreducible polynomial bases in the dataset. The binary examples are custom small-field models analogous in shape to NIST binary curves, rather than NIST standardized parameters.

A separate 14-fixture coverage extension adds degree-53 and degree-79 binary cases plus the other three prime fields. The combined 69 polynomial fixtures cover all seven core fields; the main complexity battery retains its original 55 cases and measurements.

These small fields are experimental benchmarks. An approximately 80-bit prime subgroup admits generic attacks of roughly 2^{40} group operations; this corpus makes no production-security recommendation.

The polynomial experiments

Index-calculus relation finding, with fixed two-summand targets

Each case seeks actual selected-subgroup points P and Q satisfying P+Q=T. The target is fixed before solving. Its x-coordinate is substituted into $f_3(x_1,x_2,x_3)$, and all coefficients, variables, monomial order, finite domains, and preprocessing choices are recorded as canonical sparse input.

Semaev equations used

For $y^2=x^3+Ax+B$ in odd characteristic:

$$f_3 = (x_1-x_2)^2\, x_3^2 - 2[(x_1+x_2)(x_1x_2+A)+2B]\, x_3 + (x_1x_2-A)^2 - 4B(x_1+x_2)$$

For $y^2+xy=x^3+ax^2+b$ in characteristic two:

$$f_3 = (x_1x_2+x_1x_3+x_2x_3)^2 + x_1x_2x_3 + b$$

The binary system expands every extension-field coordinate over F_2. A factor-base subspace of dimension d produces 2d Boolean variables. Semaev-derived equations are Boolean-normalized, while $v^2-v=0$ field equations remain explicit. With a fixed target the normalized f_3 coordinates have degree at most two; subgroup-membership indicators can have higher degree and must be retained in the input-degree statistics.

Main 55-fixture complexity grid

Cases	Factor bases and target cohorts
49 binary	K83-a0, S83-a1, B83-a0; low monomials d=2–6 and recorded subgroup-seeded linear bases d=2–5
6 prime	p83-S-0; 4, 8, or 16 factor-base x values from small generator multiples; exact split membership polynomials
25 planted	Constructed from eligible factor-base points; exercise recovery, not random-target success probability
30 seeded subgroup	Independently sampled reproducible nonzero subgroup targets; includes 5 empty-factor-base diagnostics

Low-monomial B83-a0 bases are empty after the selected-subgroup filter at every tested d=2–6. They remain useful unit-ideal diagnostics. Subgroup-seeded bases explicitly include eligible generator x values; their construction is biased and has its own result stratum.

A polynomial root is only a candidate. The runner lifts it to the recorded rational points, verifies membership in the chosen cyclic subgroup, and checks the exact signed sum T. It retains ordered signed decompositions and their swap orbits, allowing repeated factors. Characteristic-two polynomial coefficients do not turn −P into P in the odd-order subgroup.

The 14 added coverage fixtures use K53-a1 and K79-a1 with seeded dimension-2/3 bases, and p61-G-0, p64-G-0, and p83s-G-0 with four-x-value bases. Each configuration has a planted and a sampled target. They extend field coverage rather than the complexity grid.

What is measured

Separate yield, presentation degree, solver work, and elapsed time

Quantity	Definition in this battery
Target success	Targets with at least one verified relation / all attempted targets; a lower bound when solves are censored
Relations per attempt	Verified distinct unordered signed decompositions / all attempted targets; may exceed one
Final basis degree	Maximum total degree among polynomials in the computed final Gröbner basis
Degree of regularity	First d where the degree-d component of S/J is zero; J is generated by the highest-degree homogeneous parts of the exact recorded affine inputs
Solving degree	Maximum degree reached during the solver computation; unmeasured (null), because SymPy does not expose it
Solve time	Prebuilt-equation parsing/setup + Gröbner basis + finite-domain root extraction + independent subgroup/point verification

Timing boundary

The runs solve prebuilt fixtures. Factor-base and target construction, Semaev construction, Weil restriction, and golden-truth enumeration occur before the benchmark. Their generation time was not recorded. Both fixture_generation_seconds and generation_inclusive_total_pipeline_seconds are null; the reported solve time is not a generation-inclusive attack time.

A separate observed subprocess wall interval includes Python/SymPy startup and the optional homogeneous-ideal regularity diagnostic. Each case has a 2.5-second subprocess budget. A completed solve remains completed if only its later diagnostic times out. An unfinished solve is censored, with unknown relation yield, rather than an empty mathematical relation set.

Why the degree labels matter

Regularity here is an invariant of the recorded generator presentation. Membership filters, field equations, and preprocessing affect J. An affine unit ideal can therefore have final basis degree zero and a positive homogeneous-input regularity. Neither number measures the maximum intermediate degree reached by the solver.

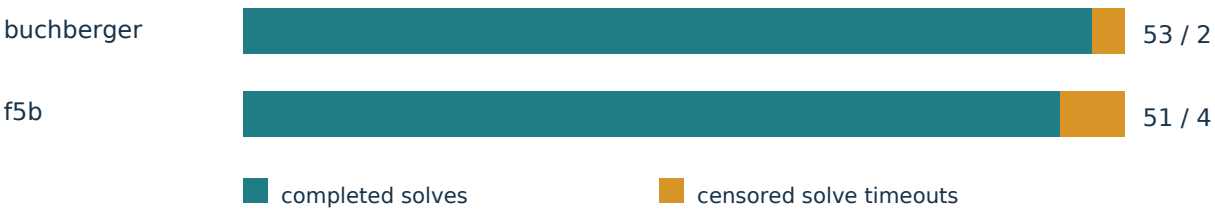
Buchberger example	Input degree	Final GB degree	Regularity
K83-a0, low d=2, planted	2	1	2
S83-a1, low d=5, planted	4	2	6
p83-S-0, base 4, planted	4	3	5
p83-S-0, base 4, sampled	4	0 (unit ideal)	5

Roots are extracted by evaluating the computed basis over the declared finite grid. The baseline uses actual Gröbner computation, followed by an explicit enumeration stage; it makes no fast triangular/FGLM-root-solver claim. Golden answers are compared only after solver roots have been obtained.

Measured solver baseline

SymPy 1.14.0 • grevlex • 2.5 seconds per case • Linux x86_64, Python 3.12.14

Method	Complete / 55	Solve timeouts	Diagnostic-only timeouts	Suite wall (s)	Σ solved (s)
buchberger	53	2	1	45.84	19.71
f5b	51	4	10	60.69	22.12



Yield by target population

Method / cohort	Done	Success / attempts	Success bounds	Unordered relations	Relations / attempt
buchberger / planted	24/25	24/25	96.0–100.0%	45	1.800
buchberger / sampled	29/30	0/30	0.0–3.3%	0	0.000
f5b / planted	22/25	22/25	88.0–100.0%	41	1.640
f5b / sampled	29/30	0/30	0.0–3.3%	0	0.000

Success bounds allow all unfinished targets to succeed for the upper endpoint. Relation counts and relations per attempt are observed lower bounds when timeouts exist. No finite upper bound on the unknown relation counts is claimed in this table.

Buchberger recovered 89 ordered / 45 unordered signed relations; f5b recovered 81 / 41. All completed cases matched the independently enumerated golden roots and signed point decompositions, with zero solver errors. No completed sampled target yielded a relation at these small factor-base sizes; the unfinished sampled target in each run remains unknown.

The method comparison describes this fixed fixture set, implementation, machine, and deadline. It does not establish a general ranking of Gröbner algorithms. The sampled cohort also includes five empty-base diagnostics; basis construction and model variant remain separate strata in the JSON rather than being interpreted as an unbiased cryptanalytic population.

Separate 14-fixture field-coverage batch

Method	Done / 14	Solve timeouts	Unordered relations	Σ solved (s)
buchberger	14	0	10	0.804
f5b	14	0	10	0.828

Each method recovered all seven planted targets; all seven sampled targets completed without a relation. Both coverage batches had zero errors or diagnostic timeouts. These are separate runs of the added fixtures, preserving the original 55-case timings above.

Isogenies and two-term maps

Explicit known answers are distinguished from class-level metadata

Map battery	Known answers	Scope
Separable degree-2 isogeny	20	E: $y^2=x^3-x$ to E′: $y^2=x^3+4x$; kernel {O,(0,0)}
Gaussian unit	40	$\varphi(x,y)=(-x, i\,y)$, where i lies in the base field
Eisenstein unit	78	$\varphi(x,y)=(\zeta_3 x,y)$, where ζ_3 lies in the base field
Relative binary Frobenius	70	$(x,y) \mapsto (x^2,y^2)$, with the coefficient-squared target model

The explicit 2-isogeny has $x'=x-1/x$ and $y'=y(1+1/x^2)$ away from its kernel. The kernel and infinity map to infinity; exceptional inputs are tested. A class identified by equal trace can lack an explicit map in this release. The dataset records this distinction and does not supply a complete isogeny graph or volcano.

Selected-subgroup endomorphism tests

The classifier supports 30 of the 56 core curves, including 6 binary Koblitz Frobenius cases. All 480 scalar decompositions verify a kernel-lattice determinant, the exact congruence $k \equiv k_1+k_2\lambda \pmod r$, nearest-plane bounds, and the signed point reconstruction $[k]G=[k_1]G+[k_2]\varphi(G)$. Unsupported cases carry explicit reasons.

For binary Koblitz coefficients in F_2, $\tau(x,y)=(x^2,y^2)$ is an endomorphism of the same curve, with $\tau^2-(2a-1)\tau+[2]=0$. For binary coefficients outside F_2, this squaring map ordinarily lands on a different coefficient-squared curve. It must not be used as a same-curve GLV map.

A map must preserve the chosen cyclic subgroup before a subgroup eigenvalue or scalar decomposition is valid. The rank-two examples test this directly: $[r]P=O$ alone cannot establish $P \in \square G \square$. Powers of an endomorphism with a quadratic minimal polynomial do not by themselves supply a balanced four-term GLV/GLS construction.

Reference timing samples

Curve	Scalar baseline (µs)	Full two-term path (µs)
p61-E3-5	247.0	191.4
p64-E4-2	174.9	161.8
K83-a0	6305.1	4338.8

Medians of three variable-time Python samples, one operation per sample. The full path includes decomposition, the endomorphism, and joint signed multiplication; precomputed joint-multiplication timings are separately named in JSON. These samples illustrate the timing interface and are separate from polynomial relation-search performance.

A genuine GLV + GLS4 example

A separate quadratic-extension supplement; the 56-curve corpus is preserved

Parameter	Verified value
Base prime / field	$p=2^{41}-853=2199023254699$; $\mathbb{F}_{p^2}=\mathbb{F}_p[u]/(u^2-2)$
Field cardinality	$q=4835703274706983025580601$ (82 bits)
Curve	$E: y^2=x^3+(7+5u)$; ordinary, $j=0$
Rational group	$C_2 \times C_{2r}$; $N=4r$
Prime subgroup	$r=120892581867617635107885 7$ (80 bits)
Geometric End ring	$\mathbb{Z}[\zeta_3]$, maximal quadratic order of discriminant -3

Let $\eta=1+u$. The commuting efficient maps are $\varphi(x,y)=(\zeta_3 x,y)$ and $\psi(x,y)=(\eta^{1-p}x^p, \eta^{3(1-p)/2}y^p)$. The subgroup eigenvalues are $\lambda_\varphi=174560118547837207691337$ and $\lambda_\psi=896867993705181372003749$. On geometric endomorphisms $\psi^2=-\pi q$; on $\mathbb{F}_{p^2}$-rational points πq acts as identity, so ψ^2 acts as $[-1]$. These are different statements.

The four-term congruence uses eigenweights 1, λ_φ, λ_ψ, $\lambda_\varphi\lambda_\psi$. The supplied reduced rank-four lattice has determinant $-r$. Exact rational Gram–Schmidt and Babai nearest-plane decomposition establish universal absolute component ceilings:

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[754746, 745967, 754745, 745966]
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Every component is therefore at most 20 bits in absolute magnitude. The accompanying two-term GLV lattice guarantees 40-bit components. Sixteen scalar vectors verify both congruences and actual point reconstructions; the supplement also records field/map answers, certified prime factors, and 51 independent PARI checks.

Four-dimensional refers to a four-term congruence lattice using these GLV and GLS maps. The geometric endomorphism algebra remains quadratic. This supplemental curve has its own scalar/map benchmark family; it is not one of the current 55 polynomial fixtures.

What remains before a full index-calculus attack benchmark

This release measures two-summand relation solving on finite factor bases. A full ECDLP pipeline additionally needs larger-arity construction, generation-inclusive timing, relation-matrix rank, linear algebra modulo r, and individual-log recovery. An instrumented solver must provide maximum working degree before a solving-degree comparison is possible. The Sage/Singular adapter exports actual equations, but no unavailable external-backend timing is fabricated.

References: I. Semaev, Summation polynomials and the discrete logarithm problem on elliptic curves, ePrint 2004/031, eprint.iacr.org/2004/031. P. Longa and F. Sica, Four-dimensional Gallant–Lambert–Vanstone scalar multiplication, ePrint 2011/608, eprint.iacr.org/2011/608. Detailed contracts: docs/ALGEBRA_METRICS.md, docs/BACKEND_PROTOCOL.md, and the canonical JSON artifacts in this release.